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RESEARCH ARTICLE

Joint Trajectory and Power Optimization for UAV-SAR Based ISAC System

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ABSTRACT Integrated Sensing and Communication (ISAC) for unmanned aerial vehicles (UAVs) has emerged as a promising paradigm for next-generation wireless systems. In this paper, we address the joint problem of resource allocation and trajectory planning for a UAV platform that integrates synthetic aperture radar (SAR) imaging and communication capabilities. The UAV operates at a fixed altitude, performing high-resolution SAR imaging of designated target areas in spotlight mode while simultaneously transmitting the collected data to a ground data center in real time. Given that the total energy consumption significantly impacts the overall performance of the UAV-SAR based ISAC system, we reformulate the trajectory planning task as a non-convex optimization problem focused on selecting optimal entry points into predefined sensing areas, with the objective of minimizing the UAV's total energy expenditure. To solve this challenging problem, we propose an efficient optimization algorithm that achieves a high-quality suboptimal solution. Extensive simulations validate the effectiveness of the proposed hybrid algorithm, which incorporates Genetic Algorithm (GA) for speed planning. It achieves faster convergence than the conventional GA-ACO method and outperforms Particle Swarm Optimization (PSO) with 3.74% reduction in total system energy consumption.

INDEX TERMS Integrated sensing and communication (ISAC), unmanned aerial vehicle (UAV), synthetic aperture radar (SAR), beamforming, trajectory optimization.

I. INTRODUCTION

In recent years, the rapid advancement of intelligent network technologies, such as autonomous driving, has created an increasing demand for the execution of communication and sensing tasks. As a result, Integrated Sensing and Communication (ISAC) has emerged as a key technology for next-generation wireless systems [1], [2]. Unlike traditional techniques that treat communication and sensing as separate functions, ISAC share spectrum and hardware resources, significantly enhancing both spectral efficiency and hardware utilization [3].

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Unmanned aerial vehicles (UAVs), as versatile airborne platforms, have been widely adopted in various applications such as ground imaging, wireless communications, and disaster monitoring [4], [5]. Due to their low cost, high mobility, and flexible deployment, UAVs have been recognized as a key technology for future ISAC systems [6], [7]. In addition to serving as aerial base stations (BS) for ground communication support, UAVs are also capable of providing radar sensing of areas. Equipped with synthetic aperture radar (SAR), UAVs can conduct high-resolution imaging of designated areas, providing a valuable complement to satellite-based SAR systems due to their enhanced maneuverability and closer proximity to target scenes [8], [9].

However, traditional UAV-SAR systems typically rely on separate hardware modules for communication and

sensing, which increase significant constraints on the UAV platform in terms of payload weight, energy consumption, and physical space [10]. ISAC offers a promising solution by integrating communication and sensing functionalities into a unified system. This integration not only reduces the hardware burden and power consumption of UAV platforms but also enables the simultaneous execution of SAR imaging and real-time data transmission to ground data centers. Consequently, UAV-SAR based ISAC systems can achieve enhanced operational efficiency and system performance.

A. RELATED WORKS

In this subsection, we focus on the combination of UAV-SAR and ISAC system. UAV-SAR has attracted significant attention due to its ability to avoid terrain obstacles and deliver higher resolution imaging compared to satellite-SAR systems.

Combined with ISAC ability to reduce payload burden through hardware and spectrum sharing, UAVs have emerged as a promising platform for integrated sensing and communication tasks [11], [12], [13], [14], [15], [16]. The synergy between ISAC lightweight design and the flexible mobility of UAVs enables more efficient deployment of UAV-ISAC systems, improving energy efficiency and operational agility in dynamic aerial environments. In [11], the authors investigated the joint UAV speed and altitude scheduling problem for energy-efficient data collection along linear infrastructures, and proposed a novel SSF-ACO algorithm that significantly reduces UAV energy consumption by accounting for altitude-dependent communication ranges and speed-energy trade-offs. In [12], the authors propose a UAV-enabled ISAC system with a movable-antenna (MA) array to support low-altitude applications, where an alternating optimization framework was developed, leveraging a Particle Swarm Optimization (PSO) based method for antenna position optimization, to enhance beamforming flexibility and maximize data throughput. In [13], the authors investigate a networked ISAC framework for low-altitude economy (LAE) scenarios, in which coordinated transmit beamforming and UAV trajectory control are jointly optimized to balance communication with authorized UAVs and 3D regional sensing by leveraging alternating optimization and successive convex approximation techniques. In [14], the authors investigate a single-UAV system with a joint receiver capable of simultaneously processing radar echoes and communication signals under a shared frequency band. A CRB-based metric is proposed to quantify sensing-communication trade-offs. In [15], the authors provide a comprehensive overview of UAV-based ISAC systems, addressing challenges such as motion control, resource allocation, and interference management. In [16], the authors propose an energy-efficient ISAC framework that jointly optimizes beamforming and waveform design to minimize energy consumption while maintaining performance.

The authors of [18], [19], and [20] focus on the estimation of sensing parameters, such as target position and distance, along with the evaluation of communication performance in UAV platforms equipped with ISAC capabilities. In [18], a cooperative sensing UAV system has been proposed, in which a rotary-wing UAV hovers while performing radar-based target detection and fusing sensing information to facilitate communication. A beam-sharing scheme was introduced to jointly support sensing and communication tasks. To enhance sensing performance, the authors optimized an upper bound of the average cooperative sensing area. However, this study did not account for the impact of the UAV's motion dynamics on sensing performance, potentially limiting its applicability in more realistic operational scenarios. In [19], the authors propose an adaptive ISAC framework for rotary-wing UAVs to jointly optimize beamforming and trajectory, enabling efficient target estimation and communication with multiple ground users. In [20], the authors investigate a fixed-wing UAV communicating with multiple users while circling around a BS, simultaneously performing target detection. Taking into account both ISAC-related power consumption and UAV propulsion energy, the research aimed to maximize the overall energy efficiency (EE) of the UAV system.

The existing research primarily focus on ISAC implementation in single-UAV systems, overlooking the potential and challenges associated with multi-UAV scenarios. To further exploit the advantages of UAVs in sensing and communication, several works [21], [22], [23], [24] have investigated the performance of ISAC systems involving multiple UAVs. In [21], the authors propose a multi-UAV-assisted ISAC framework to provide both sensing and communication services for Internet of Things (IoT) nodes in the absence of a base station. The UAV were responsible for transmitting sensing information to IoT devices and forwarding it to centralized data collection centers. In [22], the author investigate a system of multiple hovering UAV that jointly provided communication services to ground users within a designated area while collaboratively performing target sensing. The system design ensured the required positioning accuracy, balancing sensing and communication tasks under spatial constraints.

The preceding studies primarily focus on UAV-enabled ISAC systems for conventional sensing tasks, such as estimating target position and velocity. However, only a limited number of studies have investigated the integration of ISAC with SAR technology to enable high-resolution target imaging [25]. In [26], the authors propose a UAV-SAR based ISAC framework that jointly optimizes trajectory, velocity, beamforming, and power allocation to maximize energy efficiency, leveraging sequential convex optimization to significantly outperform conventional PSO-based methods. In [27], the authors investigate a bistatic SAR system involving a BS and a UAV, where the UAV simultaneously received communication signals from the BS and radar echoes reflected by targets. The UAV processed the reflected

signals to generate radio/radar images during flight. However, the study did not consider the potential application of beamforming techniques to reduce the UAV's flight energy consumption. In [28], a rotary-wing UAV equipped with SAR has been studied. The UAV performed a flat scan of a square area at a fixed angle, conducting SAR imaging during flight to improve the area scanning coverage. Nonetheless, the imaging of randomly distributed targets was not addressed. In [25], the authors investigate a rotary-wing UAV-SAR based ISAC system where the UAV scanned and imaged randomly distributed targets while simultaneously communicating with the BS. To reduce flight energy consumption, the UAV's trajectory was optimized. However, only the flight propulsion energy was considered, neglecting the energy consumption associated with SAR sensing and communication. Moreover, compared to the strip scanning mode, spotlight mode is more suitable for high-precision imaging of small target areas.

B. MOTIVATION AND CONTRIBUTIONS

Although extensive research has been conducted on UAV-SAR, there remains a research gap regarding UAV-SAR based ISAC systems. This study aims at reducing total energy consumption of UAV system, fills the gaps of energy consumption of UAV-SAR based ISAC system for target imaging in spotlight mode.

The main contributions of this paper are as follows:

- We propose a trajectory planning method for UAV-SAR based ISAC system aimed at target area imaging. Under constraints of imaging quality and communication rates with beamforming, the energy-optimal UAV path planning problem in spotlight mode is reformulated as a detection area entry point selection problem. This approach enables precise determination of the energy-efficient flight entry points and corresponding flight speeds. To solve this, we develop a dedicated optimization algorithm tailored for the UAV-SAR based ISAC system.
- The formulated problem is a mixed integer non-convex program. To reduce computational complexity, it is decomposed into two subproblems: a fuzzy-space Traveling Salesman Problem (TSP) for path sequencing, and a speed-planning problem within the detection areas. Sub-algorithm 1 employs an improved Ant Colony Optimization (ACO) algorithm that alternately optimizes the UAV's fly-in angles and detection sequence, leveraging the ACO to solve the detection sequence optimization effectively.
- The speed-planning subproblem is addressed using a Genetic Algorithm (GA). Simulation results demonstrate that the proposed hybrid algorithm achieves a suboptimal UAV trajectory with faster convergence compared to the conventional GA-ACO approach. Furthermore, compared with the PSO algorithm, the proposed method reduces the total system energy consumption by 3.74%.

The remainder of this paper is organized as follows. Section II presents the energy consumption model for the UAV-SAR based ISAC target imaging system. Section III describes the proposed optimization algorithm in detail. Simulation results and performance evaluations are provided in Section IV. Finally, Section V concludes the paper with a summary of key findings and future research directions.

II. SYSTEM MODEL

In this paper, we investigate a rotary-wing UAV equipped with SAR operating in spotlight mode to detect M target regions within a designated area, as depicted in Fig. 1. The UAV maintains a constant flight altitude throughout its mission. Upon entering a designated detection area, the UAV performs high-resolution imaging of the target regions. Simultaneously, the acquired detection data are transmitted in real time to a ground data center (DC), denoted as $\mathbf{q}_{DC}$, for further analysis and processing.

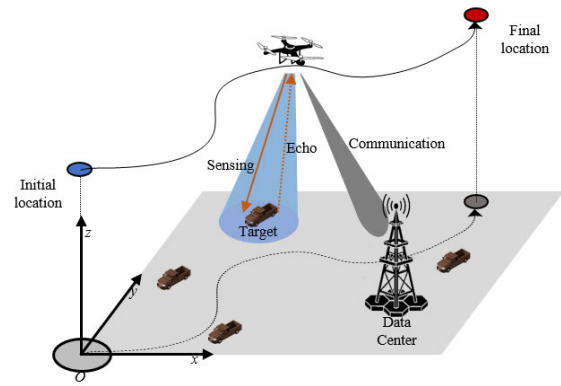


FIGURE 1. Target area imaging system of UAV-SAR based ISAC system.

The set of coordinates representing the centers of the target regions is denoted by $\mathbf{T} = \mathbf{q}_1^{\text{tar}}, \mathbf{q}_2^{\text{tar}}, \dots, \mathbf{q}_M^{\text{tar}}$. Each target region is surrounded by a circular detection area with radius R , centered at its corresponding target location. The UAV is assumed to fly at a constant altitude H , and its three-dimensional trajectory can be expressed as $\mathbf{q}(t), H$, where $\mathbf{q}(t) = [x(t), y(t)]^T \in \mathbb{R}^{2 \times 1}$ denotes the horizontal projection of the UAV's position at time t .

Given the UAV's mobility and the dual-functionality of sensing and communication, flexible beam allocation is required to ensure effective detection of target regions and reliable data transmission. During the mission, the UAV transmits SAR raw data, processed radar information, and other relevant data to the ground data center. The subsequent section focuses on the modeling and analysis of ISAC system, with particular emphasis on SAR operations and the associated UAV energy consumption.

A. UAV-ISAC MECHANISM

Considering the UAV operates in two distinct modes-flight mode and communication-sensing mode-the system's energy consumption characteristics differ accordingly. During flight mode, when the UAV travels from its initial position to

a designated detection area, energy consumption primarily arises from propulsion, as no sensing or communication tasks are performed.

Upon entering a detection area with radius $r^{(m)}$, the UAV transitions into communication-sensing mode, as illustrated in Fig. 2. In this mode, the UAV follows a straight-line trajectory within the detection region to conduct high-resolution SAR imaging, while ensuring that the resulting image satisfies the required signal-to-noise ratio (SNR) threshold. Simultaneously, it transmits the acquired data to the data center in real time, considering only the downlink. This dual-function operation reflects the integration of sensing and communication in the UAV-enabled ISAC system.

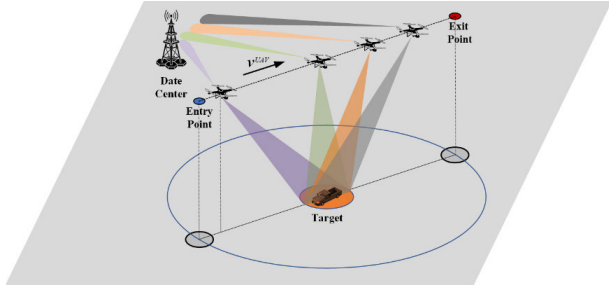


FIGURE 2. Spotlight Mode for UAV-SAR.

B. INTEGRATION OF UAV-SAR COMMUNICATION SENSING

1) RADAR SPOTLIGHT IMAGING

In this analysis, we consider a rotary-wing UAV performing SAR imaging in spotlight mode. The UAV is equipped with a $m_0 \times m_0$ planar antenna array, and the radar adopts digital beamforming (DBF) to steer the beam toward the target area. Let the transmitted SAR signal be denoted by $s_1(t)$. The elevation angle and azimuth angle from the UAV to the target are represented by ϑ_t and φ_t , respectively. The steering vector matrices at the transmitter and receiver are denoted by $\mathbf{a}_T \in \mathbb{C}^{1 \times m_0^2}$ and $\mathbf{a}_R \in \mathbb{C}^{1 \times m_0^2}$, respectively. Specifically, the transmit steering vector is defined as: $\mathbf{a}_T = [a_{11}, a_{12}, \dots, a_{1m_0}, a_{21}, \dots, a_{2m_0}, \dots, a_{m_01}, \dots, a_{m_0m_0}]$, where a_{jk} is given by: $a_{jk} = e^{j\pi[(j-1)\sin\vartheta_t \cos\varphi_t + (k-1)\sin\vartheta_t \sin\varphi_t]}$, $j, k \in [1, m_0]$. The receive steering vector is constructed as: $\mathbf{a}_R = [1, e^{j\pi \cos\vartheta_t \sin\varphi_t}, \dots, e^{j\pi(m-1)\cos\vartheta_t \sin\varphi_t}] \otimes [1, e^{j\pi \sin\vartheta_t}, \dots, e^{j\pi(m-1)\sin\vartheta_t}]$. Accordingly, the sensing channel matrix is expressed as: $\mathbf{A} \triangleq \alpha \cdot \mathbf{a}_T^H \mathbf{a}_R$, where α is denoted as the reflection coefficient. The received echo signal from the target area at the UAV is given by: $r_1(t) = \mathbf{g}_1(t)(\mathbf{A}(t)\mathbf{f}_1(t)s_1(t) + \mathbf{n}_r)$, where $\mathbf{n}_r$ denotes the stacked vector of zero-mean additive white Gaussian noise (AWGN) at the receiver input. The beamforming vectors at the transmitter and receiver are denoted by $\mathbf{f}_1 \in \mathbb{C}^{m_0^2 \times 1}$ and $\mathbf{g}_1 \in \mathbb{C}^{m_0^2 \times 1}$, respectively. The transmit power for SAR is defined as: $P^{sar} = E[\|\mathbf{a}_T s_1\|^2] = \|\mathbf{a}_T\|^2$.

We now conduct an in-depth analysis of the UAV-SAR imaging model. In practice, the 3 dB beamwidth of the radar antenna, denoted by Θ_{3dB} , is typically used to define

the effective sensing range of the radar system. Let $\theta_d(t)$ represent the instantaneous angle between the radar beam and the target at time t . Based on these parameters, the instantaneous data rate associated with the SAR perception of the target can be expressed as follows [25]:

$$R^{dr} = B_r \left(\frac{2H\Omega}{c} + \tau_p \right) \text{PRF} \text{ [bit/s]} \quad (1)$$

where c is the speed of light. The radar pulse repetition frequency is denoted as PRF. The radar bandwidth is denoted as B_r . The radar pulse duration is denoted as τ_p . And $\Omega(t)$ represents the UAV detection of the far point and the proximity of the echo distance difference at the moment t for the ratio of the UAV's flight altitude. This is calculated using the following formula:

$$\Omega(t) = \frac{\cos(\theta_1(t)) - \cos(\theta_2(t))}{\cos(\theta_1(t)) \cos(\theta_2(t))}, \quad (2)$$

where $\theta_1(t) = \theta_d(t) - \Theta_{3dB}/2$ and $\theta_2(t) = \theta_d(t) + \Theta_{3dB}/2$. Therefore difference between the far and near echo distances at moment t can be expressed as:

$$r_{far}(t) - r_{near}(t) = H \cdot \Omega(t), \quad (3)$$

the range-sensing area at time point t can be expressed as follows:

$$A_g(t) = \frac{r_{far}(t) - r_{near}(t)}{\sin\theta_2(t)} = \frac{H \cdot \Omega(t)}{\sin\theta_2(t)}, \quad (4)$$

In the Spotlight mode of operation, donate the transmit power as $P^{sar}(t)$. The signal-to-noise ratio of the image generated by the echo signal processing in the target region is as follows [26]:

$$\text{SNR}(t) = \frac{P^{sar}(t) \cdot \lambda^3 \cdot G_t \cdot G_r \cdot \sigma_0 \cdot c \cdot \tau_p \cdot \text{PRF} \cdot \cos^3\theta_d(t)}{(4\pi)^4 H^3 \cdot k \cdot T_o \cdot NF \cdot B_r \cdot L_{\text{tot}} \cdot v(t)}, \quad (5)$$

where $P^{sar}(t)$ is the transmit power of the radar and $v(t)$ is the speed of UAV at time t . The radar backscatter coefficient is denoted as σ_0 . Wavelength of radar detection is denoted as λ . G_t and G_r denote the gain coefficients for the radar antenna's transmission and reception functions, respectively. The equivalent noise factor (T_o), the Boltzmann coefficient (k), the integrated loss factor (L_{tot}), and the system noise factor (NF) are all key variables in this analysis. In practical detection scenarios, the radar imaging SNR threshold is denoted by SNR_{min} . Successful target detection can be achieved when the instantaneous SNR satisfies the condition $\text{SNR} \geq \text{SNR}_{\text{min}}$.

2) UAV TO DATA CENTER LINK

In the UAV-to-target-area communication-sensing mode, it is assumed that the UAV transmits the detected target information to the DC in real time. The non-line-of-sight (NLOS) effect is neglected, and the UAV employs DBF to steer the communication beam toward the DC. It is further

assumed that the Doppler shift induced by UAV motion can be fully compensated [31].

Under these assumptions, the received signal at the DC at time t can be modeled as follows:

$$y_{com}(t) = \mathbf{u}_1^H (\mathbf{h}(t) \mathbf{w}_1(t) s(t) + \mathbf{n}_k), \quad (6)$$

where $\mathbf{h}(t)$ denoted the communication channel matrix between UAV and DC and $\mathbf{w}_1 \in \mathbb{C}^{Q \times 1}$ and $\mathbf{u}_1 \in \mathbb{C}^{Q \times 1}$ are the transmitting and receiving beamforming vectors respectively. $s(t)$ denotes the communication signal, $\mathbf{n}_k$ denotes the additive Gaussian white noise (AWGN) at DC, which obeys the $\mathcal{CN}(0, \sigma^2)$ distribution. θ_{com} denotes the communication signal beam angle and $\theta_{com}(t) = \arctan \|\mathbf{q}(t) - \mathbf{q}_{DC}\|/H$.

Accordingly, the SNR of the communication can be calculated as follows:

$$\begin{aligned} \gamma(t) &= \frac{E[x_1^H \mathbf{w}_1^H \mathbf{h}^H \mathbf{u}_1 \mathbf{u}_1^H \mathbf{h} \mathbf{w}_1 x_1]}{E[\mathbf{n}^H \mathbf{u}_1 \mathbf{u}_1^H \mathbf{n}]} \\ &= \frac{E[\mathbf{w}_1^H \mathbf{h}^H \mathbf{u}_1 \mathbf{u}_1^H \mathbf{h} \mathbf{w}_1]}{\sigma^2 \mathbf{u}_1 \mathbf{u}_1^H}, \end{aligned} \quad (7)$$

The data rate received by the DC at moment t is calculated from Shannon's formula:

$$R_{com}(t) = B_{com} \log_2(1 + \gamma(t)), \quad (8)$$

where B_{com} denotes the communication channel bandwidth, considering the communication sensing state, the UAV forwards the detection data to the DC in real time, so the real-time communication rate of the UAV is not less than the data volume of the radar's real-time sensing of the target area, and the following conditions need to be satisfied:

$$R_{com}(t) \geq R_{min}(t), \quad (9)$$

C. UAV ENERGY MODEL

The energy consumption of UAV can be divided into two main categories: motion energy consumption and communication and sensing energy consumption of the radar. In this paper, the UAV is programmed to fly from the initial point to the target area in a sequential manner for target sensing and to return the sensing information to the DC. The two working modes of the UAV are alternated during the entire process. In this configuration, the UAV employs a flight working mode that encompasses the initial point, the first area with detection, the outpoint of the detected area, the entry point of the other detection area, and the end point. The energy consumption of this mode is primarily attributed to flight energy consumption. When the UAV reaches the target area, the UAV employs a straight-line flight path while the onboard radar operates in a spotlight mode, detecting the target area through DBF. The data is then relayed to the DC. The UAV then transitions to a communication sensing mode, which entails additional energy consumption for both flight and communication sensing.

In the majority of cases, the energy consumption of a rotary-wing UAV is primarily influenced by the flight speed and acceleration. In this study, in order to minimise the

energy loss, we attempt to reduce the energy loss caused by the UAV speed change to the greatest extent possible. Furthermore, given that the time span of a UAV speed change only represents a minor portion of the overall detection mission, it is justifiable to neglect the energy loss associated with such changes. Consequently, the UAV's flight power in relation to the real-time speed can be represented by the following equation [28]:

$$\begin{aligned} P^{\text{UAV}}(v(t)) &= P_0 \left(1 + \frac{3\|v(t)\|^2}{U_{\text{tip}}^2} \right) + C_0 \|v(t)\|^3 \\ &+ P_1 \left(\sqrt{1 + \frac{\|v(t)\|^4}{4v_0^4}} - \frac{\|v(t)\|^2}{2v_0^2} \right)^{\frac{1}{2}}, \end{aligned} \quad (10)$$

where P_1 and P_2 denote the paddle contour power and the induced power in the hovering state of the UAV, respectively, U_{tip} is the tip speed of the rotor blade, and v_0 denotes the average rotor induced speed in the hovering state. For ease of representation, $C_0 = d_f \rho s A / 2$, where d_f is the UAV fuselage drag ratio, s is the UAV rotor denseness, ρ is the air density coefficient of the UAV flight environment, and A is the UAV rotor disk area.

The flight power model of the UAV is analyzed according to Equation 10, which demonstrates that the UAV flight power is solely dependent on the flight speed. Consequently, the total energy consumption of the UAV in the single-flight operating mode can be expressed as the product of the flight power and the flight time. In order to optimize the energy consumption of the UAV, the flight path is designed to consist primarily of uniform straight lines.

Fig.3 illustrates the flight power speed curve for the UAV, which demonstrates a decrease in power followed by an increase. From the stationary to the maximum-endurance (ME) speed, the UAV power exhibits a decrease; conversely, from the maximum-endurance speed, the UAV power demonstrates an increase. It can be observed that the UAV's hovering state does not represent the lowest power consumption during flight. Consequently, the UAV does not utilize spotlight mode detection in the hovering state when identifying target areas. Given that the energy consumption of the UAV is the integral of power with respect to time, the total flight energy consumption of the UAV is not at its lowest point when the UAV is flying at the maximum-range (MR) speed in the solo flight mode. Therefore, the flight speed with the lowest total flight energy consumption of the UAV can be calculated according to the energy consumption formula, which is defined as the MR speed [32]. The UAV radar communication sensing energy consumption mainly contains the energy consumption when the airborne SAR detects the target area and the communication energy consumption when the UAV transmits the detected target data back to the UAV communication sensing power:

$$\begin{aligned} P^{\text{ISAC}}(t) &= P^{\text{com}}(t) + P^{\text{sar}}(t) \\ &= \|\mathbf{w}(t)\|^2 + \|\mathbf{a}_T(t)\|^2, \end{aligned} \quad (11)$$

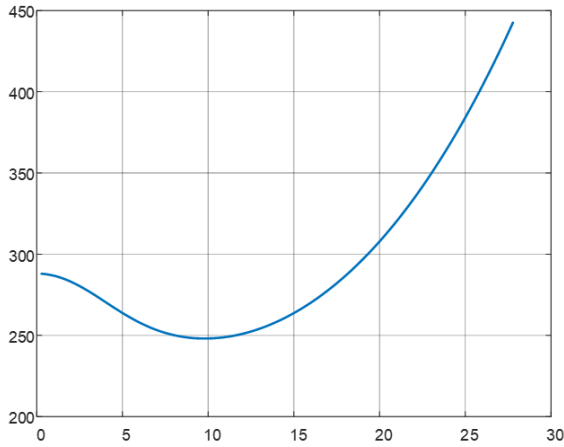


FIGURE 3. UAV Speed Power Curve.

The total power of the UAV in operation within the designated detection area is as follows:

$$P(t) = P^{\text{UAV}}(t) + P^{\text{ISAC}}(t), \quad (12)$$

D. PROBLEM MODEL AND COST FUNCTION

The UAV flies in a straight line in order to minimize the distance of the path corresponding to the minimum energy consumption. Consequently, it adopts a uniform straight line movement outside the detection area. This results in the original path planning problem being converted into the UAV detection area entry point selection problem. In light of the fact that the trajectory traversed by the UAV within the detection area is a straight line extending from the center of the target area, we may define two speeds: $\tilde{v}_{n,m,j}$, denoting the speed from the m -region to the j -region at the n -th selection, and v_{n,m,t_m} , denoting the speed at the n -th selection, at the m -region, and at the tm -th time slot. We may then define the start and end positions as $q^{(\text{in})}$ and $q^{(\text{out})}$, respectively.

Define

$$\delta_{0,i} = \begin{cases} 1, & \text{if } i = 1, \\ 0, & \text{otherwise.} \end{cases} \quad q_{0,i} = \begin{cases} q^{(\text{in})}, & \text{if } i = 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$\delta_{N+1,i} = \begin{cases} 1, & \text{if } i = 1, \\ 0, & \text{otherwise.} \end{cases} \quad q_{N+1,i} = \begin{cases} q^{(\text{out})}, & \text{if } i = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The UAV target area communication, sensing, integration, and energy minimization problem can be formulated as (13), as shown at the bottom of the next page. In light of the aforementioned discussion, the total UAV energy consumption W can be expressed as (14), as shown at the bottom of the next page and (16), as shown at the bottom of page 8.

In this context, $\mathbf{q}(t) \in \mathbf{R}^{2 \times 1}$ represents the real-time position coordinates of the UAV, while $\mathbf{q}(0)$ and $\mathbf{q}(T)$ denote the starting and end points, respectively, of the UAV's trajectory. ψ_{δ_s} , which takes on the values 0 or 1, is the state parameter. It assumes the value of 1 when the UAV transitions into communication sensing mode and 0 when it is in flight mode. Finally, $v_{\min}$, $v_{\max}$, and the speed

constraint for the UAV are also relevant factors. The speed constraint for the UAV is represented by the variable $v(t)$. The communication rate constraint for the UAV under the passive state is given by $R_{\text{com}}(t) > R^{\text{dr}}(t)$, and the communication rate of the UAV should not be smaller than the radar sensing rate to ensure real-time data transmission back to the UAV. We define the joint optimization variable set as $\chi = \{q_{n,m}^{(\text{en})}, q_{n-1,j}^{(\text{ex})}, \delta_{n,m}, \tilde{v}_{n,m,j}, v_{n,m,t}, P_{n,m,t}^{\text{sar}}, w_{n,m,t}\}$.

$$\mathbf{P1} : \min_{\{\chi\}} W \quad (15a)$$

$$s.t. \quad \sum_{n=1}^M \delta_{n,m} = \sum_{m=1}^M \delta_{n,m} = 1, \delta_{n,m} \in \{0, 1\}, \quad (15b)$$

$$v_{\min} \leq \tilde{v}_{n,m,j} \leq v_{\max}, \quad (15c)$$

$$v_{\min} \leq v_{n,m,t} \leq v_{\max}, \quad (15d)$$

$$R_{n,m}^{(\text{com})} \geq R^{(\text{tar})}, \quad (15e)$$

$$0 \leq P_{n,m}^{(\text{UAV})} \leq P_{\max}^{(\text{UAV})}, \quad (15f)$$

$$\sum_{t=1}^{T_m} v_{n,m,t} \cdot \Delta t = 2r^{(m)}, \quad (15g)$$

$$q_{n,m}^{(\text{en})} + q_{n,m}^{(\text{ex})} = 2q_m^{(\text{cen})}, \quad (15h)$$

$$\text{SNR}_{n,m}^{(\text{SAR})} \geq \text{SNR}_{\min}, \quad (15i)$$

$$n, m, j \in \{1, \dots, M\}, \quad (15j)$$

$$t \in \{1, \dots, T_m\}, \quad (15k)$$

It is imperative that the communication rate of the UAV not be less than the radar sensing rate to ensure real-time data transmission. The total power constraint of the UAV, represented by $P(t)$, is limited to $P_{\max}$, while the flight power constraint, $P^{\text{UAV}}(t)$, is constrained to $P_{\max}^{\text{UAV}}$ [33]. $\|\mathbf{w}_{\delta_s}(t)\|^2 + P_{n,m,t}^{\text{sar}}$ is the communication and sensing power constraint of the UAV; The constraint (15g), which enforces a fixed straight-line flight within the detection area, is non-convex in the context of the overall problem due to its coupling with nonlinear energy and SNR expressions. Such equality constraints are difficult to handle using standard convex optimization methods. (15i) means that the UAV will not be able to transmit the data when the UAV enters into the detection area with radius $r^{(m)}$, the UAV imaging SNR of the detection area region is not less than $\text{SNR}_{\min}$. The constraint (15i) ensures sufficient SAR imaging quality by enforcing a minimum SNR, while constraint (15e) guarantees that the radar data can be transmitted in real time via the downlink communication channel. The joint consideration of these two constraints allows the algorithm to maintain a balanced trade-off between imaging precision and communication reliability throughout the UAV's mission.

It is worth noting that the formulated optimization problem is a mixed-integer non-convex program, due to the coupling between binary decision variables and continuous variables, as well as nonlinear constraints such as radar imaging SNR and communication rate. Furthermore, the energy model and sensing constraints involve fractional and trigonometric expressions, resulting in a highly non-convex

and non-smooth objective function. The feasible space is also non-convex, owing to the discrete selection of flight paths and spotlight mode imaging geometry. These properties make conventional convex optimization methods inapplicable. To tackle this, we propose a decomposition-based heuristic algorithm that leverages ACO for trajectory planning and GA for speed optimization.

III. PROPOSED APPROACH

To solve the mixed-integer non-convex optimization problem described in Section II-D, we propose a two-stage heuristic algorithm with clear modular structure and computational tractability. The proposed approach consists of two main sub-modules: (i) a fuzzy-space TSP solved via ACO for trajectory and entry-angle planning, and (ii) a GA-based speed optimization for energy-efficient flight within detection areas. This decomposition significantly reduces the complexity of the original problem and enables fast convergence to a high-quality suboptimal solution.

Problem P1 is a notoriously difficult mixed-integer non-linear combinatorial optimization problem. To decrease the complexity of the algorithm, P1 is decomposed into two sub-problems for resolution. Namely, the trajectory planning problem regarding the selection of the entry point into the detection region and the problem of speed allocation within the detection region. Fig.4 presents the UAV-SAR based ISAC system optimization algorithm detailed flowchart. The problem is initialized by setting the normal direction from the center of the detection area to the direction of the DC as the initial entry angle. The optimization algorithm proposed herein is utilized to acquire the trajectory of the UAV and the entry points of the detection areas. Simultaneously, the GA algorithm is employed to obtain the speed allocation of the UAV within the detection areas.

Firstly, according to the two flight modes of the UAV, an analysis of the total energy consumption model of the UAV is carried out. The energy consumption issue of the UAV in the communication-sensing mode is converted into the problem of entry angle. The total energy consumption of the system is presented as (14). The total energy consumption of the UAV in each detection area is solely related to the flight speed of the UAV in each time slot, the entry point of the detection area, and the beamforming parameters. Consequently, the energy consumption of the UAV in a

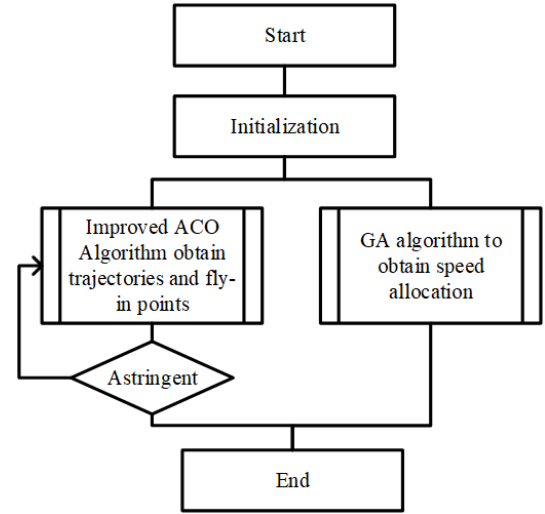


FIGURE 4. UAV-SAR based ISAC system optimization algorithm flowchart.

specific detection area is analyzed independently. The n^{th} detection area is randomly chosen, and the total energy consumption of the UAV within this area can be expressed as (13).

The velocity of the UAV corresponding to each time slot is v_t . The flight process of the UAV in the n^{th} detection area is illustrated in Fig.4. The yellow points denote the position of the UAV at the end of each time slot. Consequently, the flight distance of the UAV in the t^{th} time slot is $v_t \cdot \Delta t$. It is worth noting that based on the analysis in Section II, particularly the UAV energy cost expression in (15), the objective function is non-convex due to the presence of square roots, quartic velocity terms, and coupled nonlinearities in beamforming and propulsion models.

If the flight velocities of the UAV in each time slot are fixed, irrespective of whichever entry point the UAV selects to enter the detection area, the flight energy consumption of the UAV remains constant. The communication energy consumption within the communication sensing energy consumption $\|w_t\|^2 \cdot \Delta t$ is related to the position coordinates of the UAV at the end of each time slot. Given that the UAV flies in a straight line within the detection area, when the flight speed of each time slot is determined, the communication energy consumption is associated with the choice of the entry point. Through (10), the maximum range flight speed of the UAV can be calculated, which is the flight speed employed by the UAV in the flight mode.

$$W_n(v_t, w_t, P_t^{sar}) = \left[\sum_{t=1}^{T_n} P_1 \left(\sqrt{1 + \frac{v_t^4}{4v_0^4}} - \frac{v_t^2}{2v_0^2} \right)^{\frac{1}{2}} + P_0 + \frac{3P_0}{U_{tip}^2} \cdot v_t^2 + C_0 \cdot v_t^3 + \|w_t\|^2 + P_t^{sar} \right] \cdot \Delta t, \quad (13)$$

$$W = \sum_{n=1}^{M+1} \sum_{m=1}^M \delta_{n,m} \sum_{j \neq m}^M \delta_{n-1,j} \cdot \frac{\|q_{n,m}^{(en)} - q_{n-1,j}^{(ex)}\|}{\tilde{v}_{n,m,j}} \cdot [P^{UAV}(\tilde{v}_{n,m,j})] + \sum_{n=1}^M \sum_{m=1}^M \delta_{n,m} \cdot W_{n,m}(v_{n,m,t}, w_{n,m,t}, P_{n,m,t}^{sar}), \quad (14)$$

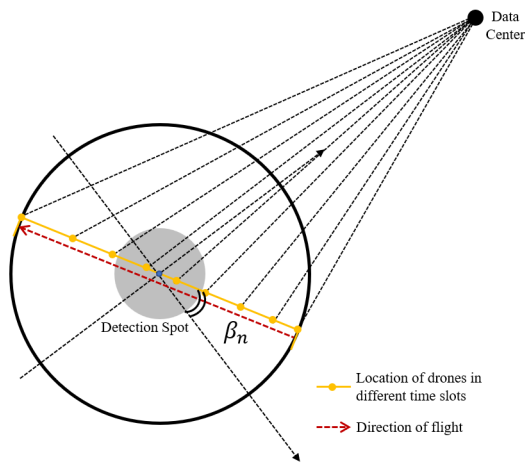


FIGURE 5. Analysis of energy consumption in the detection area.

A. UAV TRAJECTORY OPTIMIZATION

According to the analysis of the energy consumption model, the problem of UAV trajectory planning and the problem of speed distribution within the detection area are mutually independent. Solve the UAV trajectory planning issue. With the flight speed parameters in each time slot within each detection area being fixed, in the flight mode of the UAV, the total flight distance of the UAV determines the flight energy consumption. We define the joint optimization variable set as $\chi_1 = \{q_{n,m}^{(en)}, q_{n-1,j}^{(ex)}, \delta_{n,m}, P_{n,m,t}^{sar}\}$. The UAV trajectory planning problem is converted into a traveling salesman problem (TSP) in the fuzzy space and can be written as:

$$P2 : \min_{\{\chi_1\}} W_{sub1} \quad (17a)$$

$$s.t. \|q_{n,m}^{(en)} - q_m^{(cen)}\|_2 = r^{(m)}, n, m \in \{1, \dots, M\}, \quad (15b), (15f), (15i), \quad (17b)$$

where the constraint (15i), which enforces the minimum SAR imaging SNR requirement, introduces additional non-convexity into the problem. The SNR expression in (5) contains fractional and trigonometric terms, with UAV velocity appearing in the denominator, resulting in a highly nonlinear and non-convex constraint. This complexity makes (15i) difficult to handle using standard convex optimization techniques.

The direction pointing from the detection area to the data center is defined as the positive direction of the y-axis in the coordinate system, and the direction to the right perpendicular to the y-axis is the positive direction of the x-axis. A coordinate system as shown in Fig.4 is established. The rotation angle ε of the established coordinate system

relative to the standard coordinate system is:

$$\varepsilon = \arccos \frac{q_j^{(cen)} \cdot q_{DC}}{|q_j^{(cen)}| \times |q_{DC}|} + \frac{\pi}{2}, \quad (18)$$

where β_n denote the entry angle. The polar coordinate transformation of the entry point coordinate $q_{n,m}^{(en)}$ is carried out to obtain $\dot{q}_{n,m}^{(en)}$. Consequently, when the velocities in each time slot are fixed, the functional variation trend of the communication energy consumption of each detection area with respect to β_n is constant [34]. Problem P2 can be written as:

$$P3 : \min_{\{\beta_{n,m}, \delta_{n,m}\}} W_{sub1} \quad (19a)$$

$$s.t. \|\dot{q}_{n,m}^{(en)} - q_m^{(cen)}\|_2 = r^{(m)}, n, m \in \{1, \dots, M\}, \quad (15b), (15f), (15i), \quad (19b)$$

Problem P3 is a TSP problem within the fuzzy space, an expansion of the traditional TSP problem. It has been verified as an NP-hard problem, and its computational complexity will grow exponentially as the number of user groups increases. Considering that the function change trend of the communication energy consumption is fixed, the optimization direction of the entry angle for each detection area can be determined. This paper utilizes an improved UAV trajectory optimization algorithm based on the ACO algorithm to solve P3.

The ACO algorithm is an algorithm that simulates the foraging behavior of ants. When ants are moving between cities, they will leave pheromones along the paths they travel. The more concentrated the pheromones are, the more likely it is that the following ants will choose that path. As time goes by, the pheromones will slowly evaporate. At the same time, when ants find shorter routes, they will deposit more pheromones on that path, so as to lead other ants to choose better paths. When ants are choosing which detection area to visit, the probability is related to the pheromone density on the path and the weights $W(\delta_{n,m})$ between the detection areas. The weights between regions are the energy consumption of the UAV's flying and communicating. The probability of choosing between each detection area is:

$$Pro_{n,m} = \frac{\tau_{n,m}^a \cdot W(\delta_{n,m})^b}{\sum_{k \in \text{allowed}} \tau_{n,k}^a \cdot W(\delta_{n,k})^b}, \quad (20)$$

where the values a and b are the weight coefficients of pheromone and heuristic information. The *allowed* stands for the set of feasible detection areas, and τ indicates the concentration of pheromones.

$$W_{sub1}\{\chi_1\} = \sum_{n=1}^M \sum_{m=1}^M \delta_{n,m} \sum_{t=1}^{T_m} P_{n,m,t}^{sar} \cdot \Delta t + \sum_{n=1}^{M+1} \sum_{m=1}^M \delta_{n,m} \sum_{j \neq m}^M \delta_{n-1,j} \frac{\|q_{n,m}^{(en)} - (2q_j^{(cen)} - q_{n-1,j}^{(en)})\|}{v_{MR}} \times P^{UAV}(v_{MR}), \quad (16)$$

Algorithm 1 UAV Trajectory and Entry Angles Optimization

- 1: Initialize UAV motion parameters and power within bounds;
- 2: Initialize detection areas, initial entry angles of detection areas $\beta_{n,m}^{(0)}$ and DC coordinate;
- 3: set $l = 0$;
- 4: **Repeat**
- 5: Use the ACO algorithm to obtain the trajectory of the UAV and detection sequence $\delta_{n,m}^{(l)}$;
- 6: Under the condition of detection sequence $\delta_{n,m}^{(l)}$, compute values of $W^{(l)}(\delta_{n,m}^{(l)}, \beta_{n,m}^{(l)})$ and $\beta_{n,m}^{(l)}$ by convex optimization;
- 7: Set the values of $\beta_{n,m}^{(l+1)} = \beta_{n,m}^{(l)}$ as the entry points of each detection areas;
- 8: Update $l = l + 1$;
- 9: **until**
- 10: $\|W^{(l+1)}(\delta_{n,m}^{(l+1)}, \beta_{n,m}^{(l+1)}) - W^{(l)}(\delta_{n,m}^{(l)}, \beta_{n,m}^{(l)})\| \leq \varepsilon_0$ or maximum iteration reached;
- 11: **Output:** The values of minimizing UAV energy consumption $\beta_{n,m}$ and $\delta_{n,m}$.

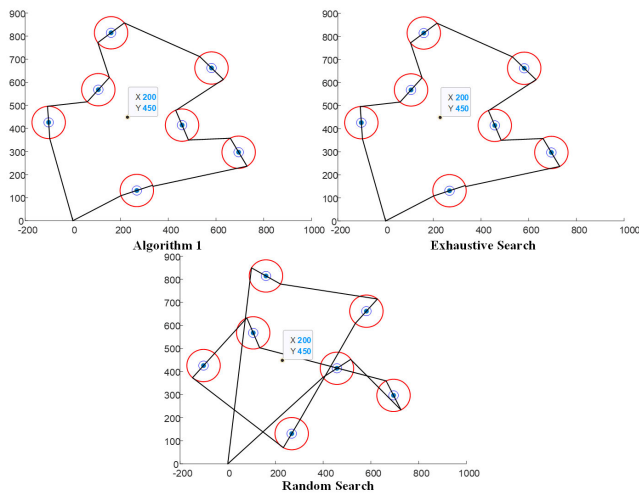


FIGURE 6. Optimized trajectory and entry angles by the algorithm 1, exhaustive search and random search, respectively.

The improved UAV trajectory optimization algorithm is optimized based on the ACO algorithm. The optimized UAV trajectory, entry angles of the detection areas, and UAV energy consumption can be obtained by alternately optimizing parameters $\delta_{n,m}$ and $\beta_{n,m}$. The details of the proposed algorithm can be referred to the Algorithm 1. Based on the analysis of the energy consumption model, the entry angles of the UAV in any detection area can be calculated and set as the initial value. The ACO algorithm is used to obtain the optimized UAV trajectory. The UAV trajectory is brought into the formula (13). Convex optimization is adopted to calculate the entry angles of each detection area and the energy consumption of UAV flight and communication. The obtained entry angles are taken as the new initial values

and iterated in turn until the results converge to obtain the optimized UAV trajectory and the entry angles of the detection areas.

As shown in Fig.2, the trajectory and entry angles optimization results obtained by algorithm 1,exhaustive search, and random search are respectively displayed.

B. UAV SPEED OPTIMIZATION IN DETECTION AREAS

Based on the above analysis, the improved UAV trajectory optimization algorithm has been utilized to acquire the UAV trajectory and the entry angles of the detection areas. Then, in problem P1, when the UAV is flying within the detection areas, there is still the problem of speed planning. Given that the total energy consumption of the UAV for detecting the target areas and flying within the detection areas is independent of the entry angles and the locations of the detection areas, this problem can be expressed as P4:

$$\mathbf{P4} : \min_{\{v_t, w_t\}} W_m \quad (21a)$$

$$s.t. \ v_{\min} \leq v_t \leq v_{\max}, \quad (21b)$$

$$\sum_{t=1}^{T_m} v_t \cdot \Delta t = 2r^{(m)}, \ t \in \{1, \dots, T_m\}, \quad (21c)$$

(15e), (15f), (15i),

where the constraint (15e), which ensures that the real-time communication rate meets the minimum requirement, also introduces non-convexity. The corresponding data rate expression in (8) depends logarithmically on the instantaneous SNR given in (7), which itself is a nonlinear fractional function of beamforming vectors and channel coefficients. As a result, constraint (15e) is non-convex and coupled with other variables, making it challenging to handle with conventional convex optimization approaches.

This paper utilizes GA algorithm to optimize P4. The GA algorithm is a random search algorithm that mimics the biological evolution process. It is a swarm intelligence algorithm with rapid convergence. Real number coding is adopted. Each chromosome represents the flight speed of the UAV in a time slot. The length of the chromosome corresponds to the total number of time slots. The fitness function directly takes the negative value of the objective function. Thus, the higher the fitness value, the better the planned speed of the UAV. The selection probability $Chri$ of the i -th chromosome is as follows:

$$Chri = \frac{-\sum_{t=1}^{T_m} (S(v_{i,t}) \cdot \Delta t)}{\sum_{j=1}^N \left(-\sum_{t=1}^{T_m} S(v_{j,t}) \cdot \Delta t \right)}, \quad (22)$$

The detailed steps of UAV speed planning using the GA algorithm are as shown in Algorithm 2. According to the preset crossover probability, the operations of selection, crossover, and mutation are continuously repeated to give rise to a new population. In each round of iteration, the

next generation population is generated on the basis of the current population. When the fitness value converges, the chromosome with the highest fitness in the population is output as the optimal UAV speed planning scheme.

$$S(v_t) = P_0 + \frac{3P_0}{U_{tip}^2} \cdot v_t^2 + P_1 \left(\sqrt{1 + \frac{v_t^4}{4v_0^4}} - \frac{v_t^2}{2v_0^2} \right)^{\frac{1}{2}} + C_0 \cdot v_t^3 + \|w_t\|^2, \quad (23)$$

Algorithm 2 UAV Speed Planning Based on the GA Algorithm

- 1: Initialize UAV motion parameters, SAR power within bounds, Chr_i , number of slots T_m , detection area diameter $2r^{(m)}$.
 - 2: Initialize population P ;
 - 3: **for** $i = 1$ to $Size$ **do**
 - 4: Create chromosome;
 - 5: **end for**
 - 6: **for** $generation = 1$ to G **do**
 - 7: Calculate Chr_i and the $S(v_t)$ for each chromosome in P ;
 - 8: Create new population P' ;
 - 9: **for** $i = 1$ to $Size$ **do**
 - 10: Select parents and add to P' ;
 - 11: **end for**
 - 12: **for** $i = 0$ to $Size - 1$ step 2 **do**
 - 13: Crossover parents in P' ;
 - 14: Mutate chromosome;
 - 15: **end for**
 - 16: $P = P'$;
 - 17: **end for**
 - 18: Find best chromosome;
 - 19: **Output:** The values of UAV speed planning.
-

C. ALGORITHMIC COMPLEXITY ANALYSIS

In this subsection, the proposed algorithm is analyzed in terms of its time complexity and space complexity respectively.

First, perform complexity analysis on Algorithm 1. Within the given upper and lower bounds, the detection order and entry angle are optimized alternately. An initial population is randomly generated, which includes the number of ants Ant , relevant parameters of the UAV, and the number of target areas n . Hence, it contains $n + 2$ path segments. At this point, the time complexity is $O(Ant(n + 2))$. Then, for each iteration, the time complexity of each ant is $O(n + 2)$. Consequently, the time complexity of all ants constructing paths is $O(Ant(n + 2))$. If T iterations are carried out, the total time complexity is $O(T * Ant(n + 2))$. After obtaining the UAV detection order, the entry angles of n areas are obtained according to the gradient descent method. At this time, the time complexity is $O(n)$. By comparing the current energy consumption with the initialized energy consumption and executing the loop

until convergence. After K cycles, the algorithm complexity is $O(K(T * Ant(n + 2) + n))$.

Next, conduct space complexity analysis on Algorithm 1. Considering the number of target areas n , $O((n + 2)^2)$ is needed. Since the number of ants is m , the space complexity at this time is $O((n + 2)^2 + Ant(n + 2))$. For obtaining the entry angle, the space complexity is $O(n)$. Considering the alternating iterative convergence of two parameters K , the space complexity of Algorithm 1 is $O(K((n + 2)^2 + Ant(n + 2) + n))$, approximately $O(Kn^2)$.

For the complexity analysis of Algorithm 2, Algorithm 2 is GA algorithm. For the speed planning of T_m time slots within the detection area and with $size$ chromosomes, the time complexity of algorithm initialization is $O(size * T_m)$. For fitness calculation and selection in each generation, the time complexities of the algorithm are respectively $O(size * T_m)$ and $O(size)$. The time complexities of crossover and mutation are $O(size * CR)$ and $O(size * MR)$ respectively. The algorithm converges after G generations. Hence, the total time complexity of the algorithm is $O(G * size * T_m)$. Regarding the space complexity analysis of Algorithm 2, considering that each chromosome has T_m time slots, the overall space complexity is $O(size * T_m)$.

Therefore, the time complexity of the UAV-SAR based ISAC System Optimization Algorithm proposed in this paper depends on the magnitudes of $K(T * Ant(n + 2) + n)$ and $G * size * T_m$. Thus, the complexity of the proposed algorithm is $O(\max(K(T * Ant(n + 2) + n), G * size * T_m))$. The algorithm space complexity depends on the magnitudes of Kn^2 and $size * T_m$. The total space is $O(\max(Kn^2, size * T_m))$.

The proposed framework exhibits good scalability with respect to the number of sensing areas. Due to the decomposition of the original problem into two loosely coupled subproblems, the overall computational complexity grows linearly with the number of detection regions, as shown in Table 2. In particular, the ACO-based trajectory planning and the GA-based speed optimization operate independently on modular subsets of variables. This allows the algorithm to maintain tractability even under extended missions involving 10 or more sensing regions.

IV. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed UAV-SAR based ISAC system optimization algorithm under realistic simulation settings. The key parameters, such as UAV altitude, detection radius, radar bandwidth, SNR threshold, and communication power limits are adopted from commonly used values in the literatures [25], [28], and [32], ensuring practical relevance. We compare our method against two baseline algorithms: the PSO algorithm and a hybrid GA-ACO approach. The primary performance metric is the total energy consumption of the UAV system, reflecting both propulsion and ISAC operation energy. Additionally, convergence and sensitivity to key parameters, such as detection area count and SNR thresholds, are analyzed for comprehensive performance evaluation.

TABLE 1. Units for magnetic properties.

Symbol	Parameters	Value
$r^{(m)}$	Radius of detection area	70m
H	UAV flight altitude	200m
PRF	Pulse Repetition Frequency	5kHz
Br	Radar bandwidth	200MHz
SNR_{min}	SAR imaging SNR threshold	20dB
V_{Max}	Maximum UAV speed	50m/s
τ_p	Radar pulse duration	$1\mu s$
B_{com}	Transmission bandwidth	200MHz
λ	Wavelength	0.1m
P_0	Blade profile power	3.4W
P_1	Rotor induced power	118W
U_{tip}	Tip speed	60m/s
v_0	Rotor velocity	5.4m/s
S	Rotor solidity	0.02
A	Disc area	0.5 m^2
$P_{max}^{(ISAC)}$	ISAC power upper limit	56dBm

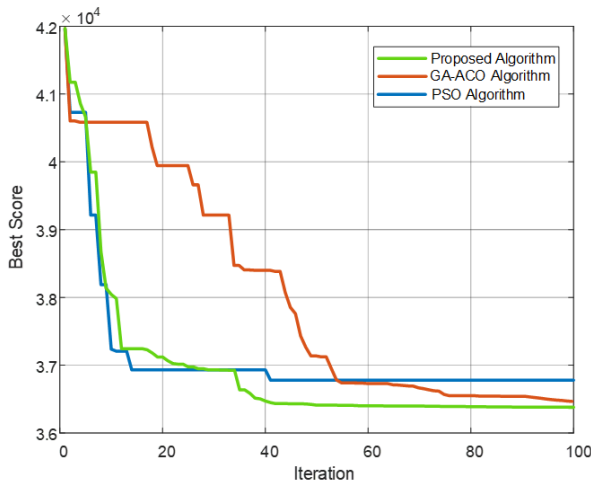


FIGURE 7. UAV-SAR based ISAC system optimization algorithm convergence.

In the simulation, we assume that the 6 targets distribute in a square area of $1\text{Km} \times 1\text{Km}$. The coordinates of DC is $[500, 550]$. The UAV starts from $[0, 0]$ to conduct SAR imaging on each target and then flies back to $[0, 0]$. The UAV flight altitude of the UAV is set as 200 m. The population size of particles is set to 100. The detailed index parameters are shown in Table 1.

As shown in Fig. 7, it can be observed that the proposed algorithm has better convergence results compared to the PSO algorithm. Normalization of energy consumption, the total UAV energy consumption of the proposed algorithm is improved by 3.74% compared to the PSO algorithm after 1000 groups of experiments are conducted independently. The proposed algorithm converges approximately after 45 iterations. The GA-ACO algorithm converges after about 75 iterations.

Conduct a comparison of the complexity between the proposed algorithm and the PSO algorithm and GA-ACO algorithm. The algorithm complexity is shown in Table 2.

TABLE 2. Algorithm complexity comparison.

Algorithm	Time Complexity	Space Complexity
Proposed Algorithm	$O(\max(K(T * Ant(n+2) + n), G * size * T_m))$	$O(\max(Kn^2, size * T_m))$
PSO Algorithm	$O(T_{ps0}N(T_m n + 2T_m + 2))$	$O(N(T_m n + 2T_m + 2))$
GA-ACO Algorithm	$O((G * size + T * Ant)T_m * n)$	$O((size + Ant) * T_m n)$

K is donated as the iteration times of the proposed algorithm. T is donated as the iteration times of the ACO algorithm in Algorithm 1. Ant is donated as the number of ants. n is donated as the number of detection areas. G is donated as the number of GA generations. $size$ is donated as the number of chromosomes. T_m is donated as the number of time slots. T_{ps0} is donated as the iteration number of the PSO algorithm. N is donated as the number of particles.

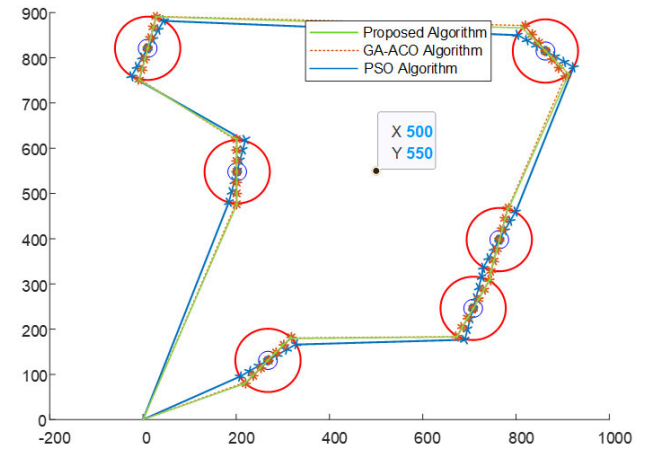


FIGURE 8. UAV ISAC trajectory planning and speed planning.

Generally, when the number of particles in the proposed algorithm, the GA-ACO algorithm, and the PSO algorithm is identical, Ant is equal to T_{ps0} , with the increase in the number of detection areas. The complexity of Algorithm 1 is higher than that of Algorithm 2, the complexity of the proposed algorithm is lower than the GA-ACO algorithm. In comparison the proposed algorithm has lower computational complexity and better convergence.

Fig.8 presents the optimized UAV trajectories and speed distributions of three algorithms. The red circles signify the detection areas, and the black dots represent the position of the DC. The green line indicates the optimized result of the proposed algorithm, the orange line represents the optimized outcome of the GA-ACO algorithm, and the blue line denotes the optimized result of the PSO algorithm. In all trajectory plans, the UAV initially enters the detection area with the speed scheme that is optimal in terms of energy consumption. Once the resolution requirements are satisfied, it flies at an optimized speed. Inside the detection area circle, each discrete point represents the distance that the UAV travels at the optimized speed in one time slot.

Subsequently, an in-depth investigation and analysis will be carried out on the minimum energy consumption of the UAV. This analysis will focus on three aspects: the quantity of

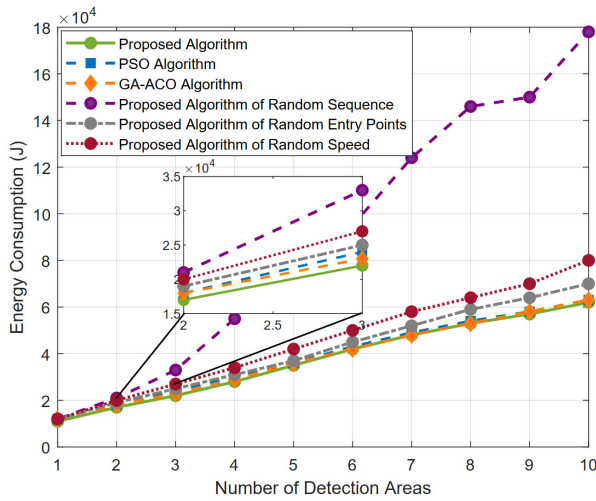


FIGURE 9. Energy Consumption vs Number of Detection Areas.

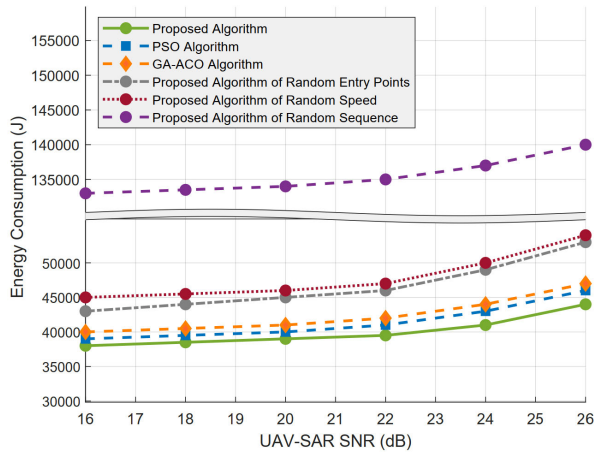


FIGURE 10. Energy Consumption vs UAV-SAR SNR.

detection regions, the received SNR of the UAV-SAR, and the communication SNR of the UAV. By systematically exploring the relationships between the minimum energy consumption and these factors, a more comprehensive understanding of the UAV's energy utilization characteristics can be achieved, which is crucial for optimizing its performance and operational efficiency in various application scenarios.

The dotted line in Fig.9 signifies the outcome of the minimum energy consumption achieved by the proposed algorithm under the aforementioned conditions. Calculate the average energy consumption of 1000 groups of detection sequences for target areas, entry points of those target areas and random flight speed parameters of the target areas within the detection area were randomly chosen. Compared with the minimize energy consumption of the proposed algorithm, the effectiveness of the proposed algorithm is demonstrated.

In a square region with dimensions of 1000×1000 units, we randomly generates 10 fixed target zones. The alteration in the overall energy consumption of the UAV is examined

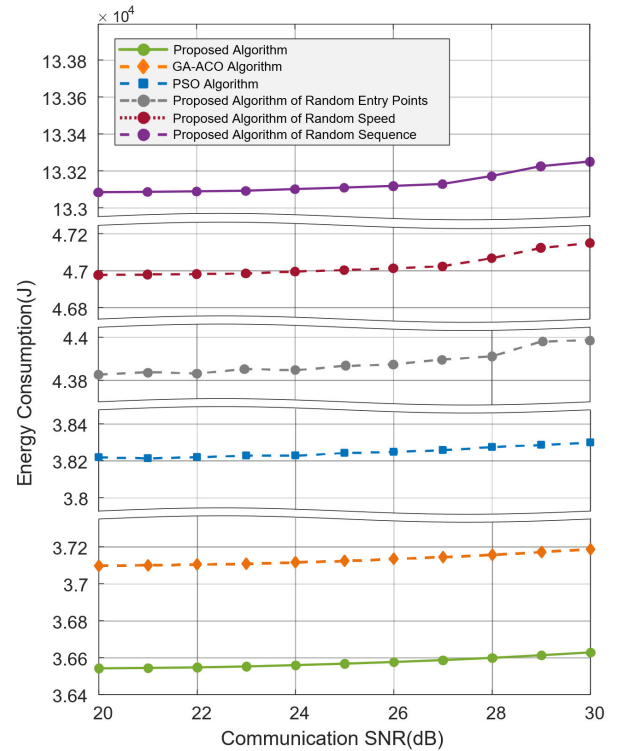


FIGURE 11. Energy Consumption vs Communication SNR.

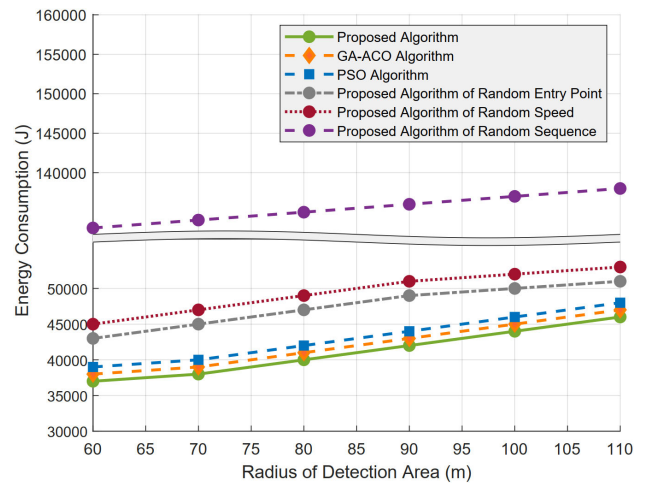


FIGURE 12. Energy Consumption vs Radius of detection areas.

as the quantity of target zones escalates from 1 to 10. It is evident from Fig.9 that, as the number of detection zones augments, the energy consumption of the UAV correspondingly increases. The simulation result shows that the order in which the regions are randomly detected by the UAV exerts a highly significant influence on the overall energy consumption of the UAV. In contrast, the entry points located within the detection areas demonstrate a comparatively lesser effect on the total energy expenditure. This disparity in the impact of these two factors highlights the importance of carefully considering

the detection sequence in the design and operation of UAV, as it can potentially lead to substantial energy savings and enhanced operational efficiency. Through a comparative analysis of the outcomes of the three algorithms, it can be deduced that the proposed algorithm demonstrates the lowest level of energy consumption, thereby presenting a more efficient solution in terms of energy utilization within this context. Despite the increasing task load, the proposed algorithm maintains stable performance and demonstrates the lowest energy consumption among all compared methods, confirming its scalability and efficiency for larger-scale missions.

Fig. 10 depicts the curve illustrating the alteration of the UAV's total energy consumption in relation to the received SNR of UAV-SAR. Upon inspection, it is discernible that the trends of variation in the total energy consumption corresponding to the random detection sequence, the random selection of entry points, and the random flight speed within the detection areas exhibit a high degree of consistency.

Under the condition of having six fixed detection areas and with a fixed SNR threshold for SAR imaging, Fig. 11 illustrates the variation curve of the total energy consumption of the UAV in relation to the SNR of UAV communication. Notably, despite the fact that the influence of the UAV communication SNR on the overall energy consumption is relatively limited, a discernible upward trend in the total energy consumption can be observed as the SNR increases. It is evident that the variation trends of the total energy consumption with respect to the random detection sequence, the random selection of entry points, and the random flight speed within the detection area exhibit a remarkable consistency.

Fig. 12 shows the curve depicting the variation of the UAV's energy consumption in relation to the radius of the detection area. It is evident from the graph that as the radius of the detection area expands, the overall energy consumption of the UAV exhibits an upward trend.

V. CONCLUSION AND FUTURE WORK

In this paper, we investigate a UAV-SAR based ISAC system for target area imaging, real-time communication, and energy-efficient trajectory planning. A joint optimization framework is proposed to simultaneously optimize the UAV's trajectory, speed, and communication-aware beamforming, with the objective of minimizing total energy consumption. The proposed algorithm demonstrates superior performance, achieving lower energy consumption than the PSO benchmark and faster convergence compared to the GA-ACO method. Although the current algorithm is designed for offline planning, its modular decomposition and rapid convergence indicate strong potential for near real-time implementation. Future work will focus on developing adaptive re-planning strategies under dynamic environments to enable real-time decision-making in response to target emergence, channel variations, or UAV perturbations.

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